

Determinant-Two Liouville Pullback

An Exact Shift-Two Frame for Literal Affine Möbius Pairs and the Growing-Progression Barrier

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Abstract

Consider positive affine forms

$$D(z) = d + sz, \quad V(z) = u + az, \quad su - ad = 2,$$

on one literal TPC generic interval. On the primitive branches exported in TPC-93, a, s are positive, coprime, and odd. The integer

$$n = sV(z) = aD(z) + 2 = su + asz$$

therefore gives an order-preserving bijection from the affine parameter to one residue class modulo as . Complete multiplicativity of the Liouville function gives the exact identity

$$\mu(D(z))\mu(V(z)) = \mu^2(D(z))\mu^2(V(z))\lambda(as)\lambda(n-2)\lambda(n).$$

We insert this identity into the full literal TPC-108 coefficient, retaining its coprimality mask, periodic factor, physical weight, additive phase, interval origin, and outer key. Thus the determinant-two signed problem is exactly a shift-two Liouville correlation on a *growing arithmetic progression*, with two quotient-squarefree masks. The pullback itself has no exponent loss and preserves every prefix. It does not turn the block into an unrestricted correlation, and known fixed-form logarithmic theorems do not supply the required growing-modulus, ordinary-weight, all-prefix estimate. The algebra is L0; its provenance-preserving attachment is L1; no L2 cancellation is proved.

1 Literal determinant-two input

Fix an interval $I \subset \mathbb{Z}$. Assume throughout that

$$D(z) = d + sz > 0, \quad V(z) = u + az > 0 \quad (z \in I), \tag{1}$$

where $a, s \in \mathbb{N}$ and

$$su - ad = 2. \tag{2}$$

The resolved nonempty branches of TPC-93 additionally satisfy

$$(a, s) = 1, \quad (as, 2) = 1, \quad (d, s) = (u, a) = 1 \tag{3}$$

[4]. In particular, a and s are odd. The positivity convention in (1) avoids extending λ to signed integers. A branch on which a form changes sign must first be partitioned at its zero; this is an exact finite operation and is not an arithmetic estimate.

For one TPC-108 key ξ , the literal inner coefficient is

$$\begin{aligned} A_{\xi, X}(z) &= \mu(D(z))\mu(V(z))\mathbf{1}_{(B, V(z))=1} \\ &\quad \times \chi(\tau + Bz)W_X(\tau + Bz), \end{aligned} \tag{4}$$

and its generic transform is

$$S_{\xi,X}(\alpha) = \sum_{z \in I} A_{\xi,X}(z) e^{-2\pi i \alpha z}. \quad (5)$$

All symbols in (4) are literal: B, τ , the fixed-period factor χ , the physical multiplier W_X , and the interval are not replaced by averaged surrogates [2].

2 The exact shift-two frame

Theorem 2.1 (Determinant-two pullback). *Under (1)–(3), put*

$$q = as, \quad n_0 = su = ad + 2, \quad n(z) = n_0 + qz. \quad (6)$$

Then:

(i)

$$n(z) = sV(z) = aD(z) + 2;$$

(ii) $z \mapsto n(z)$ is an order-preserving bijection from I onto

$$\mathcal{N}_I = \{n_0 + qz : z \in I\} = \{n \in n(I) : n \equiv 0 \pmod{s}, n \equiv 2 \pmod{a}\}; \quad (7)$$

(iii) on this progression,

$$D(z) = \frac{n(z) - 2}{a}, \quad V(z) = \frac{n(z)}{s};$$

(iv) for every $z \in I$,

$$\boxed{\mu(D(z))\mu(V(z)) = \lambda(q)\lambda(n(z) - 2)\lambda(n(z))\mu^2\left(\frac{n(z) - 2}{a}\right)\mu^2\left(\frac{n(z)}{s}\right)}. \quad (8)$$

Proof. The determinant equation gives

$$sV(z) - aD(z) = s(u + az) - a(d + sz) = su - ad = 2.$$

This proves (i), and (6) follows by expansion. Because $q > 0$, the map is injective and order preserving. Its image obeys the two displayed congruences. Conversely, since $(a, s) = 1$, those congruences select one residue modulo as , and the interval restriction selects exactly $n(I)$, proving (ii). Part (iii) is immediate.

For every positive integer m , $\mu(m) = \mu^2(m)\lambda(m)$. Complete multiplicativity and $\lambda(t)^2 = 1$ give

$$\begin{aligned} \lambda(D(z))\lambda(V(z)) &= \lambda(a)\lambda(s)\lambda(aD(z))\lambda(sV(z)) \\ &= \lambda(as)\lambda(n(z) - 2)\lambda(n(z)). \end{aligned}$$

Multiplying by the two squarefree indicators proves (8). $\square$

Remark 2.2 (Why $h_0 = 2$ is used literally). For determinant h_0 , the same calculation yields $sV = aD + h_0$ and a shift- h_0 Liouville pair. The present paper fixes $h_0 = 2$, and (3) then forces the physical progression modulus as to be odd. No average over shifts and no replacement of 2 by a convenient displacement occurs.

3 Arithmetic geometry specific to determinant two

The pullback also gives two exact structural checks that are useful when later decompositions are proposed.

Proposition 3.1 (Common-divisor and parity classification). *For every $z \in I$,*

$$\gcd(D(z), V(z)) \mid 2.$$

Under $(as, 2) = 1$, the two affine values have the same parity. Consequently:

- (i) if they are odd, their gcd is 1;*
- (ii) if they are even, their gcd is 2;*
- (iii) on the simultaneous squarefree support, the even-even case is impossible; hence $D(z), V(z)$ are odd and coprime, and*

$$\mu(D(z))\mu(V(z)) = \mu(D(z)V(z)). \quad (9)$$

Proof. Every common divisor divides $sV(z) - aD(z) = 2$. Since a and s are odd, reduction of the same identity modulo 2 gives $V(z) \equiv D(z) \pmod{2}$, proving the first two cases. If both values are squarefree and odd, they are coprime and multiplicativity of μ gives (9). If both were squarefree and even, write $D = 2D'$, $V = 2V'$, with D', V' odd. Dividing the determinant identity by 2 would give

$$sV' - aD' = 1.$$

The left side is even because a, s, D', V' are odd, a contradiction. $\square$

Remark 3.2 (A single sign is not a linear sign). Equation (9) compresses the active two-form sign to one Möbius value of a reducible quadratic polynomial. It does not turn it into $\mu(cz + r)$. Therefore a single-linear-form estimate cannot be inserted merely because only one μ symbol remains.

Proposition 3.3 (Ordered sequence isometry). *Let c_z be any complex sequence on I , and put $\tilde{c}_{n(z)} = c_z$ on $\mathcal{N}_I$. Then, for every $1 \leq p < \infty$,*

$$\sum_{z \in I} |c_z|^p = \sum_{n \in \mathcal{N}_I} |\tilde{c}_n|^p,$$

and the terminal maximum and endpoint-anchored variation satisfy

$$\begin{aligned} \max_{T \in I} \left| \sum_{\substack{z \in I \\ z \leq T}} c_z \right| &= \max_{M \in \mathcal{N}_I} \left| \sum_{\substack{n \in \mathcal{N}_I \\ n \leq M}} \tilde{c}_n \right|, \\ |c_{\max I}| + \sum_{z, z+1 \in I} |c_{z+1} - c_z| &= |\tilde{c}_{\max \mathcal{N}_I}| + \sum_{n, n+q \in \mathcal{N}_I} |\tilde{c}_{n+q} - \tilde{c}_n|. \end{aligned}$$

Proof. The map $z \mapsto n_0 + qz$ is an order-preserving bijection, so each identity is a relabeling of the same finite list. $\square$

The last proposition shows that the sharp signed-prefix/BV norm of TPC-122 is transported exactly [3]. Measuring variation between consecutive ambient integers instead of consecutive progression points would change the norm and is not authorized by the pullback.

4 Complete coefficient transport

Define, for $n \in \mathcal{N}_I$,

$$\begin{aligned} z(n) &= \frac{n - n_0}{q}, \\ \mathcal{Q}_{a,s}(n) &= \mu^2\left(\frac{n-2}{a}\right) \mu^2\left(\frac{n}{s}\right), \\ \mathcal{W}_{\xi,X}(n) &= \mathbf{1}_{(B,n/s)=1} \chi(\tau + Bz(n)) W_X(\tau + Bz(n)). \end{aligned} \tag{10}$$

Corollary 4.1 (Literal block in the shift-two frame). *The transform (5) has the exact representation*

$$S_{\xi,X}(\alpha) = \lambda(q) \sum_{n \in \mathcal{N}_I} \mathcal{Q}_{a,s}(n) \lambda(n-2) \lambda(n) \mathcal{W}_{\xi,X}(n) e^{-2\pi i \alpha z(n)}. \tag{11}$$

Every initial z -prefix corresponds to the same initial n -prefix, and

$$\sum_{z \in I} |A_{\xi,X}(z)| = \sum_{n \in \mathcal{N}_I} \mathcal{Q}_{a,s}(n) |\mathcal{W}_{\xi,X}(n)|. \tag{12}$$

Proof. Insert (8) in (4)–(5). The mask and all remaining factors become (10). The map in [theorem 2.1](#) is an order-preserving bijection, which proves both the prefix assertion and (12). $\square$

Remark 4.2 (The phase is not silently simplified). One may write

$$e^{-2\pi i \alpha z(n)} = e^{-2\pi i \alpha (n - n_0)/q}$$

on $\mathcal{N}_I$. The quotient is integral there. We retain $z(n)$ in (11) so that no choice of a representative for α/q changes the stated physical phase.

5 What the pullback does and does not prove

Proposition 5.1 (No algebraic exponent loss). *Let*

$$V_{\xi,X} = \sum_{z \in I} |A_{\xi,X}(z)|.$$

Any estimate proved for the right side of (11) in its displayed progression, masks, weight, phase, and all-prefix quantifiers,

$$|S_{\xi,X}(\alpha)| \leq X^{-\eta+o(1)} V_{\xi,X},$$

returns the same exponent η to the original block.

Proof. The two sums are equal and their absolute comparison masses agree by (12). $\square$

Proposition 5.2 (Growing-progression firewall). *The identity (11) does not by itself reduce the required estimate to an unrestricted sum $\sum_{n \leq y} \lambda(n-2) \lambda(n)$. A valid returning theorem must retain at least:*

- (a) the actual, generally growing modulus $q = as$ and residue $n_0 \bmod q$;
- (b) both quotient-squarefree masks $\mathcal{Q}_{a,s}$;
- (c) the actual interval origin and every required prefix;
- (d) the coprimality and periodic masks in $\mathcal{W}_{\xi,X}$;
- (e) the physical weight and additive phase; and

(f) *uniformity over the predeclared literal outer family.*

Proof. Each item occurs explicitly in (11). Dropping one changes either the support or the coefficient. Equality to an unrestricted correlation would already fail for the constant coefficient when $q > 1$, since only one residue class is sampled. $\square$

The logarithmically averaged two-point theorem applies to fixed affine forms with reciprocal weight and its own terminal-window quantifiers [1]. It is not an ordinary-weight, power-rate theorem uniform in the growing data of [proposition 5.2](#). This observation is a type check, not a negative statement about that theorem.

6 A finite determinant-two example

Take

$$a = 3, \quad s = 5, \quad d = u = 1.$$

Then $su - ad = 2$, $q = 15$, and

$$D(z) = 1 + 5z, \quad V(z) = 1 + 3z, \quad n(z) = 5 + 15z.$$

The first few values illustrate both exact zeros and surviving signs:

z	n	D	V	$\mu(D)\mu(V)$
0	5	1	1	1
1	20	6	4	0
2	35	11	7	1
3	50	16	10	0
4	65	21	13	-1

For example, at $z = 4$,

$$\lambda(15)\lambda(63)\lambda(65) = -1 = \mu(21)\mu(13),$$

and both quotient-squarefree masks equal 1. At $z = 1$, the Liouville product is still a sign, but the quotient-squarefree mask vanishes because $V = 4$. This is why that mask cannot be dropped after the shift-two sign has been exposed.

7 Stop/go certificate

For a pulled-back block let $N = |I|$ and $q = as$.

GO. Continue through this frame only if a later theorem is stated on the actual progression and controls all literal masks, weights, phases, origins, and prefixes. Because [proposition 5.1](#) is exact, a proved power saving then returns without an additional algebraic loss.

SHORT STOP. If N is below the predeclared long-block threshold, the arithmetic-progression reformulation creates no new averaging. The block remains in the short-fiber ledger.

UNIFORMITY STOP. If the modulus, mask, or interval range lies outside a cited theorem's uniformity, that theorem is not callable. One may not replace q by a fixed modulus or the progression by all integers.

ARITHMETIC STOP. The statement

$$\max_{T \in I} \left| \sum_{\substack{n \in \mathcal{N}_I \\ z(n) \leq T}} \mathcal{Q}_{a,s}(n) \lambda(n-2) \lambda(n) \mathcal{W}_{\xi,X}(n) e^{-2\pi i \alpha z(n)} \right| \ll X^{-\eta} V_{\xi,X}$$

with a fixed $\eta > 0$ is a new fixed-shift arithmetic input. It is not proved here and must be labelled L2 if established.

8 Claim boundary

Statement	Level and status
Determinant-two n -frame and identity (8)	Exact finite algebra (L0).
Order, prefix, phase, mask, and mass preservation	Exact finite algebra (L0).
Attachment to every verified TPC-93/TPC-108 primitive branch	Literal fixed-2 interface (L1).
Uniform growing-progression cancellation	Not proved; new arithmetic input (L2 if proved).
H3, positive zero-mode exponent, or physical endpoint	Not proved.

TPC-127 gives a canonical arithmetic coordinate for the signed determinant-two block. It is a strict improvement in definition: the remaining sign is now the literal shift-two Liouville product, but on a sparse growing progression and behind quotient-squarefree masks. The next finite task is to expand those masks with their actual congruence and tail costs. No parity breakthrough, prime-pair lower bound, Hardy–Littlewood asymptotic, or twin-prime conclusion is claimed.

References

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